# Worst-pair term: paired comparison

Pre-registered in `COMPARE_worstpair_PROTOCOL.md`. Generated 2026-10-20T19:19:01.878Z · git 6d96721. Exploratory: placeholder wire and helmet; these configurations only.

A = solver without the term (lambdaMax 0). B = with the smooth term (lambdaMax 1). Primary outcome: worst nearest-ring |k|.

- 1: B lower in >= 25/35: **met**
- 2: median ratio <= 0.90: **met**
- 3: no coverage drop below 0.99: **met**

B lower in 26 / 35 runs · median ratio B/A 0.838 · coverage violations 0. **Decision: adopt B as default**

| set | knot | N | seed | worst A | worst B | B/A | ≥thr A | ≥thr B | median A | median B | 2nd-ring max A | 2nd-ring max B | cov A | cov B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| fast | T(2,3) b=0.4 | 16 | 1 | 0.0102 | 0.0159 | 1.55 | 3/29 | 2/31 | 0.0024 | 0.0030 | 0.067 | 0.068 | 0.999 | 0.998 |
| fast | T(2,3) b=0.4 | 16 | 2 | 0.0330 | 0.0134 | 0.41 | 4/31 | 3/31 | 0.0031 | 0.0050 | 0.065 | 0.075 | 0.999 | 0.998 |
| fast | T(2,3) b=0.4 | 16 | 3 | 0.0424 | 0.0343 | 0.81 | 7/32 | 20/30 | 0.0058 | 0.0119 | 0.067 | 0.078 | 1.000 | 1.000 |
| fast | T(2,3) b=0.4 | 16 | 4 | 0.0395 | 0.0207 | 0.53 | 4/31 | 7/31 | 0.0038 | 0.0044 | 0.070 | 0.064 | 0.999 | 0.998 |
| fast | T(2,3) b=0.4 | 16 | 5 | 0.0362 | 0.0172 | 0.48 | 8/30 | 4/30 | 0.0045 | 0.0042 | 0.061 | 0.072 | 0.999 | 0.998 |
| fast | T(2,3) b=0.4 | 32 | 1 | 0.0543 | 0.0368 | 0.68 | 14/69 | 40/71 | 0.0045 | 0.0115 | 0.060 | 0.060 | 0.999 | 0.999 |
| fast | T(2,3) b=0.4 | 32 | 2 | 0.0385 | 0.0389 | 1.01 | 19/67 | 38/71 | 0.0046 | 0.0107 | 0.062 | 0.059 | 1.000 | 0.999 |
| fast | T(2,3) b=0.4 | 32 | 3 | 0.0470 | 0.0363 | 0.77 | 17/68 | 32/70 | 0.0044 | 0.0099 | 0.061 | 0.064 | 1.000 | 0.999 |
| fast | T(2,3) b=0.4 | 32 | 4 | 0.0413 | 0.0359 | 0.87 | 25/73 | 34/69 | 0.0055 | 0.0099 | 0.060 | 0.058 | 0.999 | 0.999 |
| fast | T(2,3) b=0.4 | 32 | 5 | 0.0367 | 0.0395 | 1.08 | 16/69 | 42/72 | 0.0046 | 0.0124 | 0.062 | 0.066 | 1.000 | 1.000 |
| fast | T(2,5) b=0.4 | 16 | 1 | 0.0335 | 0.0329 | 0.98 | 5/30 | 8/30 | 0.0031 | 0.0049 | 0.059 | 0.059 | 0.999 | 1.000 |
| fast | T(2,5) b=0.4 | 16 | 2 | 0.0327 | 0.0416 | 1.27 | 3/30 | 26/33 | 0.0030 | 0.0190 | 0.058 | 0.058 | 0.999 | 0.999 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| fast | T(2,5) b=0.4 | 16 | 3 | 0.0334 | 0.0339 | 1.02 | 8/30 | 8/30 | 0.0068 | 0.0047 | 0.052 | 0.055 | 0.999 | 0.998 |
| fast | T(2,5) b=0.4 | 16 | 4 | 0.0324 | 0.0313 | 0.97 | 9/30 | 9/30 | 0.0052 | 0.0061 | 0.061 | 0.059 | 0.999 | 0.998 |
| fast | T(2,5) b=0.4 | 16 | 5 | 0.0374 | 0.0317 | 0.85 | 9/30 | 8/30 | 0.0064 | 0.0070 | 0.053 | 0.057 | 1.000 | 1.000 |
| fast | T(2,5) b=0.4 | 32 | 1 | 0.0542 | 0.0356 | 0.66 | 25/71 | 52/71 | 0.0074 | 0.0145 | 0.052 | 0.053 | 0.999 | 1.000 |
| fast | T(2,5) b=0.4 | 32 | 2 | 0.0434 | 0.0364 | 0.84 | 25/67 | 47/69 | 0.0044 | 0.0145 | 0.053 | 0.055 | 0.999 | 0.999 |
| fast | T(2,5) b=0.4 | 32 | 3 | 0.0436 | 0.0352 | 0.81 | 27/68 | 50/70 | 0.0073 | 0.0136 | 0.053 | 0.051 | 1.000 | 0.999 |
| fast | T(2,5) b=0.4 | 32 | 4 | 0.0520 | 0.0365 | 0.70 | 32/69 | 50/70 | 0.0096 | 0.0160 | 0.053 | 0.052 | 0.999 | 1.000 |
| fast | T(2,5) b=0.4 | 32 | 5 | 0.0387 | 0.0356 | 0.92 | 17/66 | 46/68 | 0.0040 | 0.0148 | 0.057 | 0.052 | 1.000 | 1.000 |
| fast | T(3,4) b=0.3 | 16 | 1 | 0.0443 | 0.0433 | 0.98 | 17/30 | 17/28 | 0.0105 | 0.0126 | 0.077 | 0.075 | 1.000 | 0.997 |
| fast | T(3,4) b=0.3 | 16 | 2 | 0.0761 | 0.0436 | 0.57 | 18/30 | 24/30 | 0.0137 | 0.0172 | 0.076 | 0.075 | 0.999 | 0.998 |
| fast | T(3,4) b=0.3 | 16 | 3 | 0.0865 | 0.0432 | 0.50 | 17/30 | 20/30 | 0.0139 | 0.0164 | 0.077 | 0.076 | 0.998 | 0.999 |
| fast | T(3,4) b=0.3 | 16 | 4 | 0.0359 | 0.0443 | 1.24 | 18/30 | 13/28 | 0.0138 | 0.0088 | 0.075 | 0.078 | 0.999 | 0.998 |
| fast | T(3,4) b=0.3 | 16 | 5 | 0.0428 | 0.0431 | 1.01 | 20/30 | 22/30 | 0.0136 | 0.0156 | 0.077 | 0.075 | 0.999 | 0.998 |
| fast | T(3,4) b=0.3 | 32 | 1 | 0.0507 | 0.0740 | 1.46 | 34/68 | 63/73 | 0.0104 | 0.0290 | 0.078 | 0.071 | 0.997 | 0.997 |
| fast | T(3,4) b=0.3 | 32 | 2 | 0.0792 | 0.0749 | 0.95 | 37/67 | 62/71 | 0.0111 | 0.0283 | 0.074 | 0.070 | 0.999 | 0.997 |
| fast | T(3,4) b=0.3 | 32 | 3 | 0.0800 | 0.0591 | 0.74 | 34/69 | 58/72 | 0.0100 | 0.0271 | 0.073 | 0.069 | 0.997 | 0.999 |
| fast | T(3,4) b=0.3 | 32 | 4 | 0.0817 | 0.0768 | 0.94 | 42/69 | 61/71 | 0.0117 | 0.0280 | 0.071 | 0.069 | 0.999 | 0.998 |
| fast | T(3,4) b=0.3 | 32 | 5 | 0.0841 | 0.0587 | 0.70 | 40/70 | 59/71 | 0.0140 | 0.0253 | 0.073 | 0.068 | 1.000 | 0.997 |
| full | T(2,3) b=0.4 | 16 | 1 | 0.0088 | 0.0114 | 1.30 | 0/29 | 1/31 | 0.0016 | 0.0028 | 0.066 | 0.067 | 1.000 | 0.998 |
| full | T(2,3) b=0.4 | 16 | 2 | 0.0356 | 0.0096 | 0.27 | 3/31 | 0/31 | 0.0022 | 0.0035 | 0.072 | 0.077 | 0.999 | 0.998 |
| full | T(2,3) b=0.4 | 16 | 3 | 0.0425 | 0.0343 | 0.81 | 7/32 | 18/30 | 0.0051 | 0.0124 | 0.066 | 0.075 | 1.000 | 1.000 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| full | T(2,3) b=0.4 | 16 | 4 | 0.0392 | 0.0222 | 0.57 | 3/31 | 7/31 | 0.0023 | 0.0029 | 0.067 | 0.063 | 0.998 | 0.998 |
| full | T(2,3) b=0.4 | 16 | 5 | 0.0364 | 0.0168 | 0.46 | 5/30 | 4/30 | 0.0018 | 0.0015 | 0.067 | 0.077 | 0.999 | 0.999 |